

ECON 0100 | Vignette B4 | Efficiency | Solutions

Due in Recitation. Vignettes are a certificate of the work done together in Recitation.

Solution guide for the teaching team. Answers and working are in the red blocks; everything else is what students see. The working follows the handwritten guides to the Fall 2022 butterbeer price ceiling and price floor vignettes, the source of both questions: label the regions A to F, compare CS and PS before and after, and name the transfer and the DWL.

The supply and demand curves for pumpkin pasties are giving in **Vignette B3**.

Q1 | Price Ceiling

The Ministry of Magic decided to institute a price ceiling of 7 galleons to allow many more to afford the treat. Use a graph to evaluate the welfare effects this policy had on the market: label and calculate CS, PS, DWL, and which direction incentives push prices due to this policy.

CS: _____

PS: _____

DWL: _____

Shortage/Excess: _____

Direction incentives push prices: _____

CS 45 galleons. PS 25 galleons. DWL 2 galleons. A shortage of 4 pasties. Incentives push prices up.

Equilibrium. From Vignette B3, $P_d = 14 - \frac{1}{2}Q_d$ and $P = 2 + \frac{1}{2}Q_s$ meet at $Q^* = 12$, $P^* = 8$. The ceiling of 7 is below P^* , so it binds.

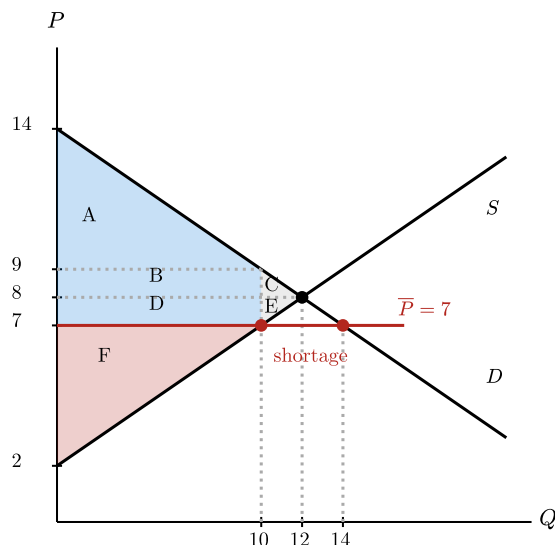
Quantities at the ceiling.

$$7 = 2 + \frac{1}{2}Q_s \rightarrow Q_s = 10$$

$$7 = 14 - \frac{1}{2}Q_d \rightarrow Q_d = 14$$

The shortage is $14 - 10 = 4$ pasties. Sellers are the short side, so only 10 pasties are exchanged. The last of them is worth $V = 14 - \frac{1}{2} \cdot 10 = 9$ galleons to its buyer.

Areas. $A = (14 - 9) \cdot 10 \cdot \frac{1}{2} = 25$, $B = (9 - 8) \cdot 10 = 10$, $C = (12 - 10) \cdot (9 - 8) \cdot \frac{1}{2} = 1$, $D = (8 - 7) \cdot 10 = 10$, $E = (12 - 10) \cdot (8 - 7) \cdot \frac{1}{2} = 1$, $F = (7 - 2) \cdot 10 \cdot \frac{1}{2} = 25$.



	Pre price ceiling	Post price ceiling
CS	$A + B + C = 36$	$A + B + D = 45$
PS	$D + E + F = 36$	$F = 25$
DWL	0	$C + E = 2$

The price ceiling lowered the price, created a shortage, and generated DWL. It also transferred welfare $D = 10$ from PS to CS. Consumers gain D but lose C , so CS rises by $10 - 1 = 9$ producers lose D and E , so PS falls by $10 + 1 = 11$.

Direction. Up. Buyers who can't get pasties would offer a little more to jump the line, and sellers would accept, but with the legal price held at 7 they can't, so the shortage persists despite the incentives to raise the price.

Q2 | Price Floor

The Ministry of Magic decided that the market price wasn't sufficiently high enough for sellers, and decided to institute a price floor of 11 galleons to raise the price. Use a graph to evaluate the welfare effects this policy had on the market.

CS: _____

PS: _____

DWL: _____

Shortage/Excess: _____

Direction incentives push prices: _____

CS 9 galleons. PS 45 galleons. DWL 18 galleons. An excess of 12 pasties. Incentives push prices down.

Equilibrium. $Q^* = 12$, $P^* = 8$, as in Q1. The floor of 11 is above P^* , so it binds.

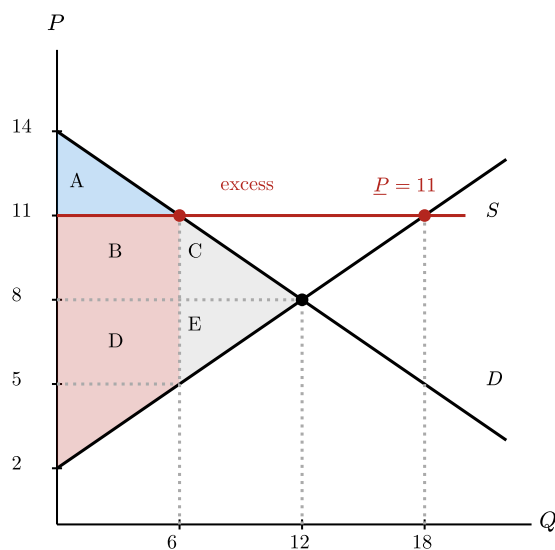
Quantities at the floor.

$$11 = 14 - \frac{1}{2}Q_d \rightarrow Q_d = 6$$

$$11 = 2 + \frac{1}{2}Q_s \rightarrow Q_s = 18$$

The excess is $18 - 6 = 12$ pasties. Buyers are the short side, so only 6 pasties are exchanged. The last of them costs $2 + \frac{1}{2} \cdot 6 = 5$ galleons to make.

Areas. $A = (14 - 11) \cdot 6 \cdot \frac{1}{2} = 9$, $B = (11 - 8) \cdot 6 = 18$, $C = (12 - 6) \cdot (11 - 8) \cdot \frac{1}{2} = 9$, $D = (8 - 5) \cdot 6 + (5 - 2) \cdot 6 \cdot \frac{1}{2} = 27$, $E = (12 - 6) \cdot (8 - 5) \cdot \frac{1}{2} = 9$.



	Pre price floor	Post price floor
CS	$A + B + C = 36$	$A = 9$

	Pre price floor	Post price floor
PS	$D + E = 36$	$B + D = 45$
DWL	0	$C + E = 18$

This policy raised the price, created excess, transferred $B = 18$ from consumers to producers, and eliminated C and E from total surplus. CS falls by $18 + 9 = 27$ PS rises by $18 - 9 = 9$.

Direction. Down. Sellers left holding unsold pasties would cut their price a little to find a buyer, and buyers would accept, but with the legal price held at 11 they can't, so the excess persists despite the incentives to lower the price.